



Smart Farm Monitor For Agricultural Production And The Environment



OVERVIEW

- Definition of smart farming
- Smart farming robot using Arduino
- Design flow
- Why we used GPS with compass ?
- Software design
- Advantages of this system
- Suggested additions and changes

DEFINITION OF SMART FARMING

- Using modern technology to increase the quantity and quality of agricultural products



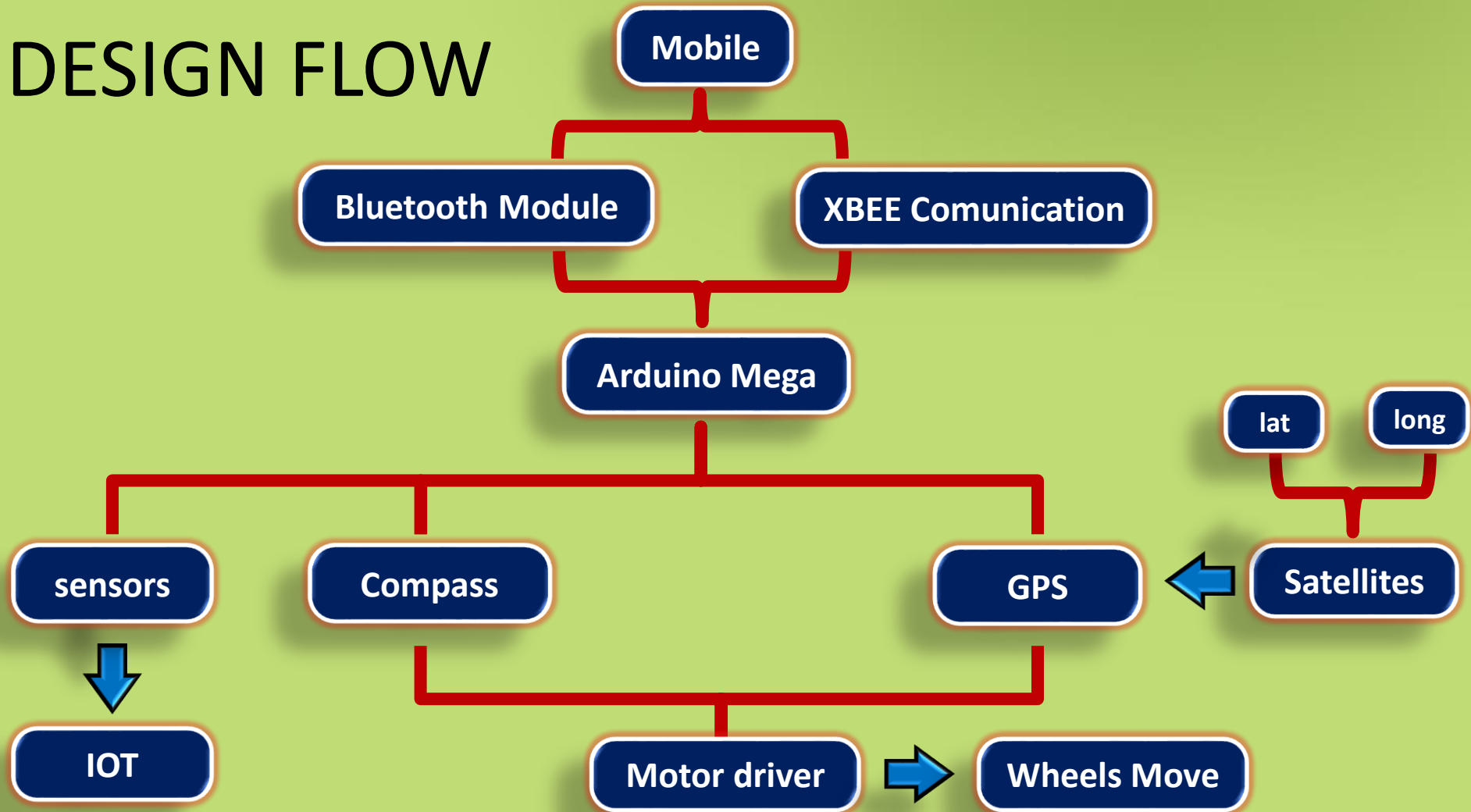


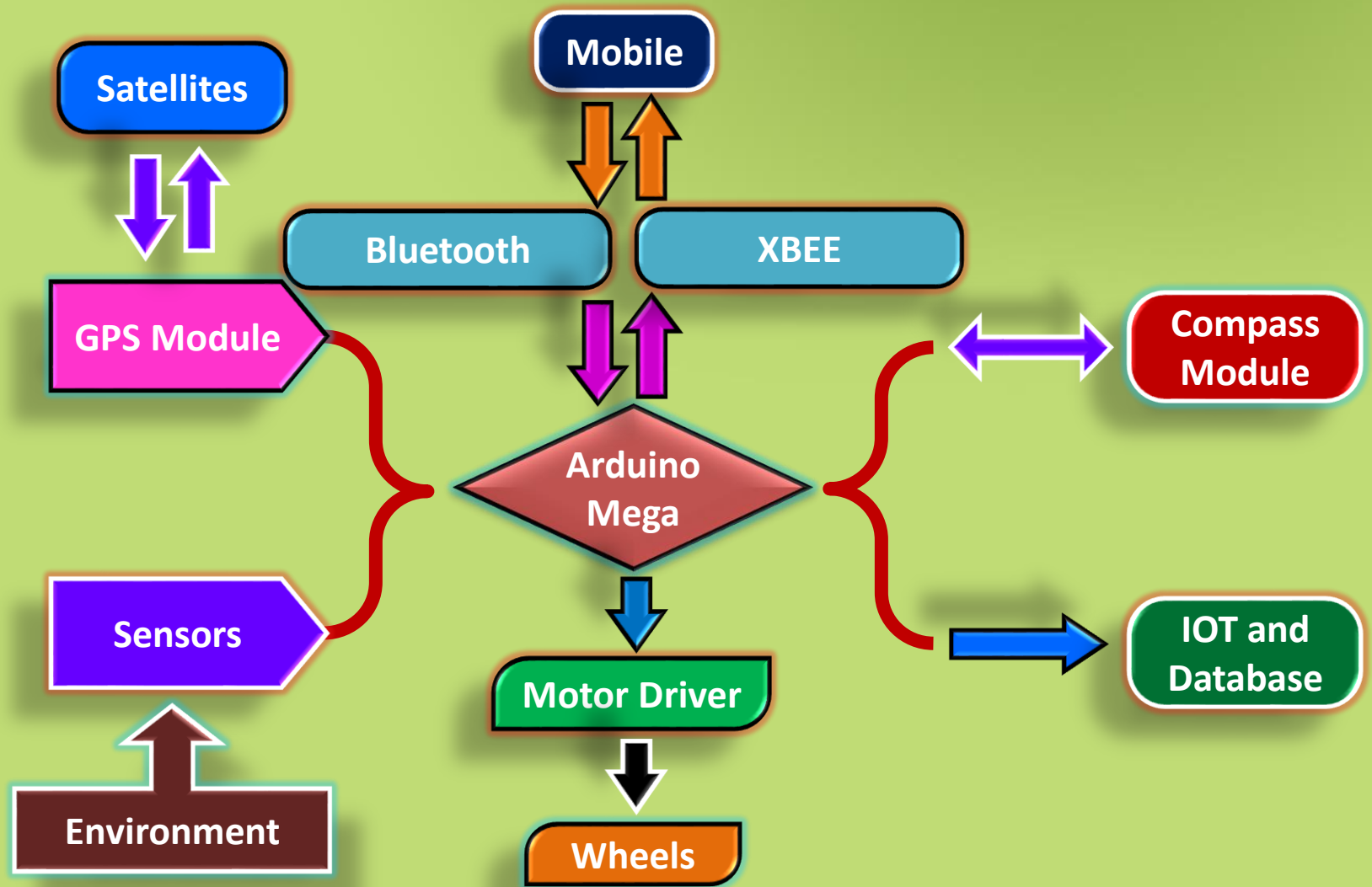
SMART FARMING ROBOT USING ARDUINO

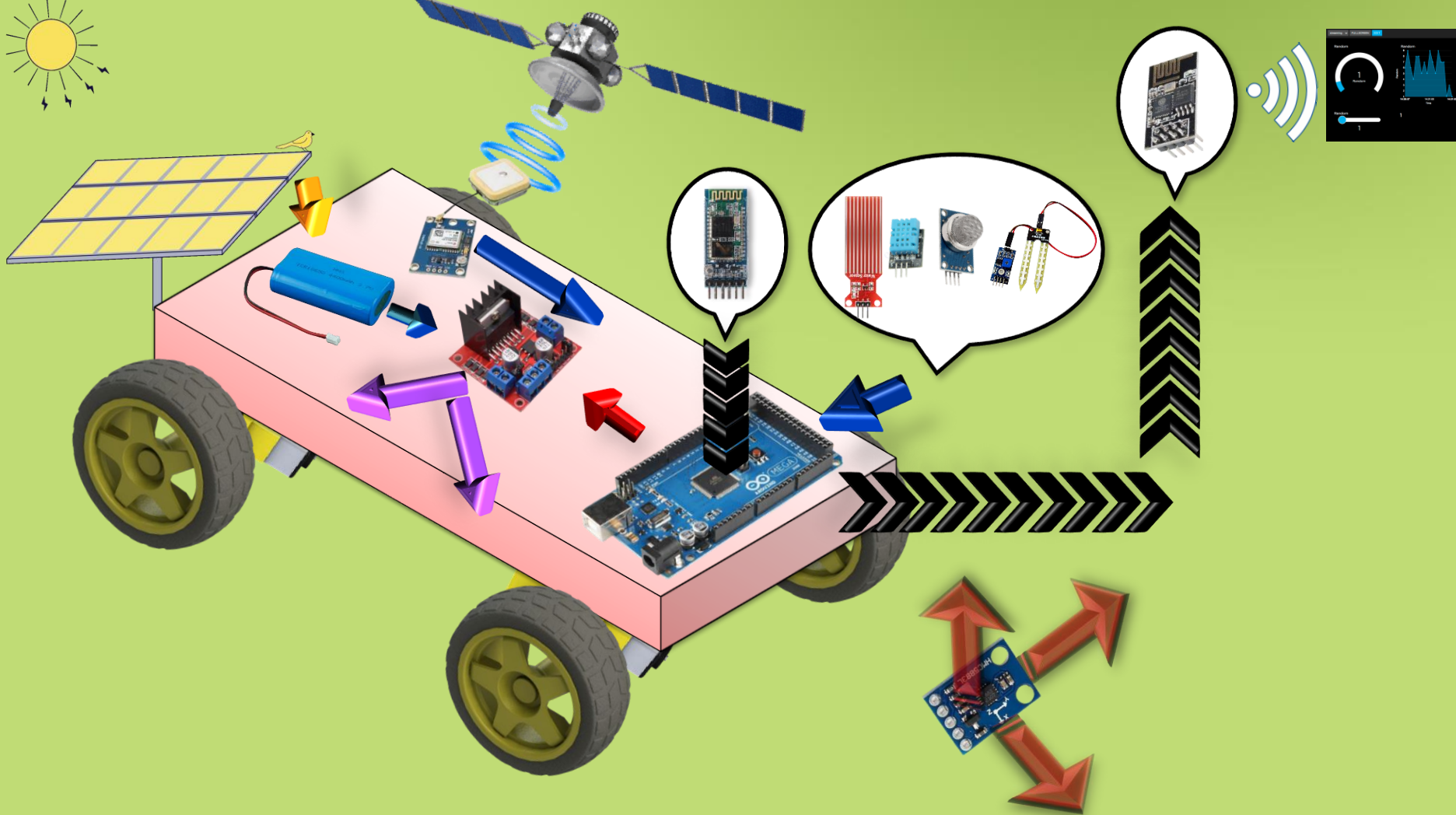
➤ Functions

- Helping the farmer take care of the plants without supervision
- Give accurate measurements of soil moisture
- Measurement of temperature, humidity and rain in the atmosphere

DESIGN FLOW







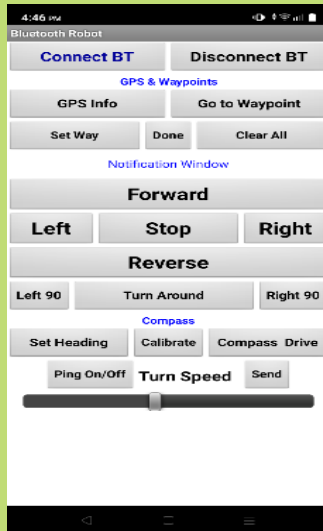


WHY WE USED GPS WITH COMPASS ?

- Outdoor
- Long distance
- It can be changed permanently
- Easy to use

SOFTWARE DESIGN

- Control robot App
- Sending Data to Computer



The screenshot shows a Microsoft Excel spreadsheet with the following data:

	A	B	C	D	E	F	G
1	Temperature	soil moisture	CO2 Level (PPM)	Rain	Date	Time	
2	20	9%	755	0%	3/12/2022	7:12PM	
3	21	9%	755	0%	3/12/2022	7:13PM	
4	20	12%	623	21%	3/12/2022	7:14PM	
5	20	10%	621	23%	3/12/2022	7:15PM	
6	20	12%	655	16%	3/12/2022	7:16PM	
7	20	12%	622	27%	3/12/2022	7:17PM	
8	19	13%	545	28%	3/12/2022	7:18PM	
9	20	13%	548	57%	3/12/2022	7:19PM	
10	21	14%	602	49%	3/12/2022	7:20PM	
11	20	15%	633	55%	3/12/2022	7:21PM	
12	21	16%	564	61%	3/12/2022	7:22PM	
13	20	16%	562	67%	3/12/2022	7:23PM	
14	20	16%	425	70%	3/12/2022	7:24PM	
15	19	17%	467	79%	3/12/2022	7:25PM	
16	20	18%	464	85%	3/12/2022	7:26PM	
17	19	18%	434	86%	3/12/2022	7:27PM	
18	20	19%	423	83%	3/12/2022	7:28PM	



ADVANTAGES OF THIS SYSTEM

- Multipurpose System
- Environmentally friendly (Clean)
- Give accurate results
- Easy to use
- Acceptable Cost



SUGGESTED ADDITIONS AND CHANGES

- Make the robot closing loop
- Add smart electronic arm to the robot
- Analyzing some data using vision (artificial intelligence) Easy to use
- Design a database to store the received data





Questions?



Thankyou